

Explanation – HW09: Linear Transformations, Kernel & Image

Part 1: Foundations – What you need to know

1. What is a linear transformation?

A **linear transformation** $T: V \rightarrow W$ is a function between vector spaces that satisfies two properties:

1. $T(u + v) = T(u) + T(v)$ (addition is respected).
2. $T(\lambda v) = \lambda T(v)$ (scalar multiplication is respected).

Analogy

A linear transformation is like a **structure-preserving translator**: if I first add two vectors and then translate, the result is the same as if I first translate each individually and then add. Example: “multiply by 2” is linear, “+5” is not (because $u + v + 5 \neq (u + 5) + (v + 5)$).

Representation matrix

Every linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be represented by an $m \times n$ matrix A : $T(v) = Av$.

Columns of A = images of the standard basis vectors: $A = (T(e_1) \mid T(e_2) \mid \cdots \mid T(e_n))$.

2. Kernel and Image

Definitions

- **Kernel**: $\text{Ker}(T) = \{v \in V : T(v) = 0\}$. “What gets mapped to zero?”
 - **Image**: $\text{Im}(T) = \{T(v) : v \in V\}$. “What can possibly come out?”
- Both are subspaces – $\text{Ker}(T) \subseteq V$ (domain), $\text{Im}(T) \subseteq W$ (codomain).

Analogy

T is a “machine”. The **kernel** is the set of “wasted inputs” (come out as 0, so information is lost). The **image** is the set of all possible outputs.

3. The Rank-Nullity Theorem

The most important theorem

For every linear transformation $T: V \rightarrow W$ with finite domain dimension:

$$\dim \text{Ker}(T) + \dim \text{Im}(T) = \dim V.$$

Analogy

If the machine has n input knobs ($\dim V = n$), then: “wasted knobs” (kernel) + “effective knobs” (image) = n . Each knob is either one or the other – nothing is counted twice.

4. Injective, Surjective, Bijective – connection to Kernel & Image

Characterizations

For $T: V \rightarrow W$:

$$\text{injective} \iff \text{Ker}(T) = \{0\} \iff \dim \text{Ker}(T) = 0.$$

$$\text{surjective} \iff \text{Im}(T) = W \iff \dim \text{Im}(T) = \dim W.$$

$$\text{bijective} \iff \text{injective and surjective} \iff T \text{ invertible}.$$

Important consequence

For $T: V \rightarrow W$ with $\dim V = \dim W$ (square matrix case):

$$\text{injective} \iff \text{surjective} \iff \text{bijective}.$$

But: if $\dim V \neq \dim W$, then T can **never be bijective**.

- $\dim V > \dim W$ (e.g. $\mathbb{R}^5 \rightarrow \mathbb{R}^3$): T cannot be injective (kernel has at least dimension $\dim V - \dim W$).
- $\dim V < \dim W$ (e.g. $\mathbb{R}^3 \rightarrow \mathbb{R}^5$): T cannot be surjective.

5. Strategy – how do I solve such problems?

Recipe

1. **Set up representation matrix** A (columns = $T(e_i)$).
2. Apply **Gaussian elimination** to A .
3. **Kernel**: solve $Av = 0$. Free variables $\rightarrow$ basis vectors of the kernel.
4. **Image**: pivot columns of the row echelon form $\rightarrow$ corresponding *original columns* form the basis.
Warning: Do not take the columns of the row echelon form, but those of the original matrix!
5. **Check Rank-Nullity Theorem**: $\dim \text{Ker} + \dim \text{Im} \stackrel{?}{=} \dim V$.
6. **Inj./Surj./Bij.**: read off directly from the dimensions.

Part 2: Exercise 35

$T: \mathbb{R}^5 \rightarrow \mathbb{R}^3$. Already from the dimensions, T can never be injective (and therefore never bijective/invertible). The only remaining questions are how “bad” the kernel is and whether T is at least surjective.

Approach: A single Gaussian elimination delivers the answers for (a), (b), (c). Pivots are at variables a, b – so c, d, e are free variables in the kernel, and columns 1, 2 of A form the basis of the image.

Concretely: $\dim \text{Ker} = 3$, $\dim \text{Im} = 2$. Rank-Nullity: $3 + 2 = 5 = \dim \mathbb{R}^5$ ✓. *Not injective* (kernel $\neq \{0\}$), *not surjective* (image $\neq \mathbb{R}^3$), *not invertible*.

(c) – the trick: Solving $T(v) = (1, 2, 5)$ = finding a particular solution v_0 , then adding the entire kernel. (Like “general solution = particular solution + homogeneous solution” for differential equations.)

Part 3: Exercise 36

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Now square – T could be invertible.

Strategy for (a)

Instead of setting up a matrix and computing the determinant: try to find T^{-1} directly. Set $(x, y, z) = T(a, b, c)$ and solve for a, b, c . If this works uniquely $\Rightarrow T$ is invertible; the solution *is* then T^{-1} .

Why does this work? “Unique back-resolution” means precisely: for every output there is exactly one input. That is bijective.

(b) and (c) follow directly from (a)

Rule of thumb

If we know that T is invertible (hence bijective), then **no further computation** is needed:

$$\text{Ker}(T) = \{0\}, \quad \text{Im}(T) = W = \text{the entire codomain.}$$

And T is automatically injective, surjective, bijective.

Part 4: Exercise 37

$T: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$, defined as “multiply on the left by M ”. Note: $\dim \mathbb{R}^{2 \times 2} = 4$.

Key observation

The matrix $M = \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix}$ is **singular** (rank 1: row 2 = 4 · row 1, column 2 = 2 · column 1).

Consequence

For $T(X) = MX$ with rank-1 matrix M :

- The product MX has, in each column, a multiple of column 1 of M (i.e. $(1, 4)^\top$).
- Therefore $\dim \text{Im}(T) = 2$ (one dimension per output column).

Direct approach

Compute $T(X)$ with general $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then read off the conditions for kernel and image. We get:

$$T(X) = \begin{pmatrix} a + 2c & b + 2d \\ 4(a + 2c) & 4(b + 2d) \end{pmatrix}.$$

From this directly:

- Kernel: $a + 2c = 0$ and $b + 2d = 0$. Two free variables (c, d) . $\dim \text{Ker} = 2$.
- Image: every matrix of the form “first row arbitrary, second row = 4 · first”. $\dim \text{Im} = 2$.

Rank-Nullity Theorem: $2 + 2 = 4$ ✓.

Part 5: Exercise 38 – Existence Questions

Main tool for existence problems

Step 1: Dimension check. Compute $\dim \text{Ker}$ and $\dim \text{Im}$ using the Rank-Nullity Theorem. Check whether the values

1. are *integers*,
2. are ≥ 0 ,
3. satisfy $\dim \text{Im} \leq \dim W$ (codomain).

If one check fails $\rightarrow$ no such T exists.

If all checks pass $\rightarrow$ construct a concrete example.

(a) $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, **Ker = Im**

$\text{Ker} = \text{Im}$ forces equal dimensions. Rank-Nullity: $2 \dim \text{Ker} = 2 \Rightarrow \dim \text{Ker} = 1$. Integer, all good $\Rightarrow$ exists.

Example: $T(a, b) = (b, 0)$ “forgets a , pushes b into the first component”. Kernel = Image = x -axis.

(b) $\mathbb{R}^3 \rightarrow \mathbb{R}^3$, **Ker = Im**

$2 \dim \text{Ker} = 3 \Rightarrow \dim \text{Ker} = \frac{3}{2}$. **Not an integer** $\Rightarrow$ does not exist.

(c) $\mathbb{R}^6 \rightarrow \mathbb{R}^3$, **$\dim \text{Ker} = 2$**

Rank-Nullity: $\dim \text{Im} = 6 - 2 = 4$. But $\text{Im} \subseteq \mathbb{R}^3$, so $\dim \text{Im} \leq 3$. **Contradiction** $\Rightarrow$ does not exist.

Recognize the pattern

Such problems are mostly pure dimension calculations – the linear transformations are only “wrapping”. Write down the three numbers $\dim V$, $\dim W$, $\dim \text{Ker}$, compute $\dim \text{Im} = \dim V - \dim \text{Ker}$, and check whether it fits into the codomain.